

Computer Engineering Department

King Mongkut's University of Technology Thonburi

CPE 241 Database

Problem Session #06 Feedback on ERD with Normalization Exercises

Date: Check PS06 assessment activities on LEB2

Introduction

In this lab, after selecting your project database, you will collaborate with two other groups working on similar topics. You will discuss your design with them, exchange feedback, and review their database structures. Based on the feedback received, you will refine your database design to improve its efficiency and structure. Alongside the first task, you will also practice normalization by working through a set of exercise problems to enhance your understanding of database optimization.

Task List

- 1. Brainstorming on your ERD**
- 2. Normalization exercises**

Task 1: Brainstorming

In this task, you will practice effectively presenting your database design using an Entity-Relationship Diagram (ERD). This activity is part of active learning and peer teaching, where you will explain your design to your peers and gain insights from their perspectives. Your goal is to ensure that other groups clearly understand your design while also learning about theirs. After presenting, you will exchange constructive feedback, providing suggestions to improve their designs and receiving recommendations to refine your own. This collaborative process will help strengthen your database modeling skills and enhance your ability to communicate technical concepts.

Objectives of Task 1

1. Find two groups that share the similar project
2. Practice communicating with ERD
3. Exchange your design and feedback
4. Improve your design

Instructions

STEP 1: Prepare Your ERD Presentation

- Review your ERD to ensure it accurately represents your database design.
- Identify key components such as entities, attributes, primary keys, foreign keys, and relationships.
- Be ready to explain the logic behind your design choices (e.g., why certain relationships exist, how normalization is applied).
- Organize your explanation so that your peers can follow your thought process easily.

STEP 2: Present Your ERD to Two Other Groups

- Each group will take turns presenting their ERD to the others.
- Clearly explain the structure and functionality of your database, including:
 - The purpose of your database.
 - The entities and how they relate to one another.
 - Any design decisions you made, such as normalization or indexing.
- Answer questions from the other groups and clarify any uncertainties.

STEP 3: Provide and Receive Feedback

- After each presentation, actively listen to other groups' designs and ask questions.
- Identify strengths and potential improvements in their database structure.
- Provide constructive feedback by suggesting:
 - Ways to improve relationships between entities.
 - Opportunities for better normalization.
 - Any missing attributes or constraints that could enhance functionality.
- Take notes on the feedback you receive and consider how to incorporate it into your design.

STEP 4: Revise Your ERD Based on Feedback

- Reflect on the feedback you received and decide which changes to implement.
- Update your ERD to improve efficiency, consistency, and usability.
- Save your revised ERD, as you will use it in later tasks.

STEP 5: Find Common Entities and Relationships

After discussing with other groups, identify and compile the common entities and relationships recognized across all groups.

Deliverables

1. Your initial ERD before receiving feedback.
2. A summary of the feedback you received from other groups.
3. Your revised ERD after incorporating suggestions.
4. A brief reflection on how the discussion helped refine your database design.
5. A list of common entities and relationships among your group and the others.

Submission Format

- Compile all written explanations and diagrams into a single PDF report.
- Organize the report with clear sections.
- Include captions for each diagram explaining PK, FK constraints, and weak entity dependencies.

Task 2: Normalization Exercise

This task is just an exercise to practice your skill in database normalization. You will be given three Unnormalized Form (UNF) of databases. Your goal is to create a physical level ERD along with SQL to create all the tables involved. You will have to apply normalization steps to the UNFs up to 3rd Normal Form (3NF). Somehow, the BCNF is a stricter kind of 3NF, it would be good to also examine the relationships whether they are already in BCNF form.

Objectives of Task 2

1. Apply normalization techniques to create relational databases
2. Identify the normal form of each table in your databases with a brief explanation.
3. Create SQL statements

Instructions

STEP 1: Normalize the Following Unnormalized Database Schemas

- Students-Courses-Grades (id, name, phone1, phone2, address 1, address 2, course name, courseDescription, teachers, grade, registerDate, examDate)
(See Table 1 in the Appendix)
- Doctor-Patient-Appointment (docName, specialty, docPhoneOffice, docPhoneMobile, patName, diagnose, threatment, insurance, hospital, building, floor, roomNumber)
(See Table 2 in the Appendix)
- Artist-Album-Song (artist, genre, album name, songs, songGenre, songLength)
(See Table 3 in the Appendix)

STEP 2: Normalize the Schemas

From UNF, you need to apply steps to normalize the schemas into BCNF or 3rd normal form. In short, removing repeated information, giving a referential key, then removing dependency.

STEP 3: Determine Constraints and Physical Data Types

In this step, you will have to consider adding some data constraints in addition to primary key and foreign key constraints. After that, you will have to think of data types of each attribute that will be stored in DBMS.

Step 4: Create SQL Statement

The final step is to translate the physical level ERD and turn it into tables in the database. That requires you to write out the **CREATE** statements for all those tables and relationships.

Deliverables

1. ERD showing the normalized entities/relationships
2. SQL statements to create tables in DBMS database.

Submission Format

- Compile all written explanations and diagrams into a single report. It must include requirement list, draft ERDs, and the justification document.
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Final Submission

- Combine results from Task 1 and Task 2 in a PDF file and ensure it is named appropriately (e.g., **PS06_YourGroupName.pdf**) then submit the report on LEB2 before the deadline.
- Upload the PDF report to the Assessment Activities section for PS06 on LEB2.

Appendix

- Table 1 Students-Courses-Grades

First Name	Last Name	Department Name	GPA	Phone 1	Phone 2	Phone 3	Email Primary	Email 2	Email 3	Course Name	Course Description	Credit	Registration Date	Grade
John	Doe	Computer Science	3.75	1234567890	0987654321	NULL	john.doe@example.com	j.doe@univ.edu	-	Database Systems	Introduction to databases	3	2024-02-17	A
Jane	Smith		3.60	2345678901	8765432109	NULL	jane.smith@example.com	-	-	Linear Algebra	Concepts of vector spaces and matrices	3	2024-02-17	B
Alice	Johnson	Computer Science	3.90	3456789012	NULL	NULL	alice.j@example.com	a.johnson@univ.edu	-	Database Systems	Introduction to databases	3	2024-02-18	A
Bob	Brown	Physics	3.50	4567890123	NULL	NULL	bob.brown@example.com	-	-	Quantum Mechanics	Study of wave-particle duality	3	2024-02-19	C

- Table 2 Doctor-Patient-Appointment

Doctor Name	Specialty	Office Phone	Mobile Phone	Patient Name	Diagnosis	Treatment	Insurance	Hospital	Building	Floor	Room Number	Appointment Date
Dr. John Smith	Cardiology	123-456-7890	987-654-3210	Alice Johnson	Hypertension	Prescribed medication...	Aetna	City General Hospital	North Wing	3	305A	2024-02-17
Dr. Emily Davis	Dermatology	234-567-8901	876-543-2109	Robert Brown	Eczema	Topical steroid cream	Blue Cross	Metro Medical Center	East Wing	2	204B	2024-02-18
Dr. John Smith	Cardiology	123-456-7890	987-654-3210	Michael Carter	High Cholesterol	Prescribed statins	United Healthcare	City General Hospital	North Wing	3	305A	2024-02-19
Dr. Emily Davis	Dermatology	234-567-8901	876-543-2109	Jessica Miller	Acne	Oral antibiotics and skincare	Cigna	Metro Medical Center	East Wing	2	204B	2024-02-20

- **Table 3 Artist-Album-Song**

Artist	Country	Genre	Album Name	Release Year	Record Label	Album Length	Song Title	Track #	Song Genre	Song Length
The Beatles	UK	Rock	Abbey Road	1969	Apple Records	00:47:23	Come Together	1	Rock	00:04:20
The Beatles	UK	Rock	Abbey Road	1969	Apple Records	00:47:23	Something	2	Rock	00:03:02
The Beatles	UK	Rock	Abbey Road	1969	Apple Records	00:47:23	Here Comes the Sun	7	Rock	00:03:06
Taylor Swift	USA	Pop	1989	2014	Big Machine	00:48:41	Blank Space	2	Pop	00:03:51
Taylor Swift	USA	Pop	1989	2014	Big Machine	00:48:41	Shake It Off	6	Pop	00:03:39
Eminem	USA	Hip-Hop	The Eminem Show	2002	Shady Records	01:17:16	Without Me	4	Hip-Hop	00:04:50
Eminem	USA	Hip-Hop	The Eminem Show	2002	Shady Records	01:17:16	Sing for the Moment	12	Hip-Hop	00:05:40